The Wishverse Under Adversarial Granting:

How Coarse Specification Breaks the Sufficiency of a Single Wish

Isaac Kaimfa

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Abstract

A *wish* is informally an operator that receives a state of the world and produces a different one. Modeled naively as a non-deterministic map $w\colon S \to \mathscr{P}(S)$, a wish is just a binary relation, and the theory reduces to abstract rewriting. We recover the missing structure by separating three ingredients the naive model conflates: *feasibility* (what a granting agent could deliver), *preference* (what the wisher wants), and *expressibility* (the language in which wishes can be phrased). The residual non-determinism of the naive map is then granter discretion, resolved adversarially against the preference: the formal monkey's paw. In this static, order-theoretic setting we first recover, as reformulations, two known impossibility results from the theory of misspecified objectives: that nontrivial betrayal-free specification is impossible over rich wish languages (an order-theoretic counterpart of the unhackability results of Skalse et al.), and that a wish language too coarse to isolate the optimum makes every expressible wish betrayable (a bounded counterpart of the incomplete-proxy result of Zhuang and Hadfield-Menell). Our new contribution is sequential. We introduce the *wishing game*, prove a single-cell reduction showing the wisher's only power is to confine the granter to one language cell, and establish the paper's central result: when feasibility is *transitive*, one wish is optimal for every utility if and only if the language is fine enough to separate each state's successors, and otherwise multi-step wishing is strictly more valuable, by the full unit-bounded margin, even though the feasibility relation is transitive. The mechanism is not intransitivity of any quotient relation, as we had conjectured and here refute, but the granter's freedom inside a coarse cell: forced to be stingy on utility, an adversarial granter may be forced to hand the wisher a better stepping stone. Betrayal and the value of sequential wishing are thus two faces of one operator inequality, the strictness of an abstract controllable-predecessor operator below its concrete counterpart, which under static transitive feasibility and a fixed observation partition makes coarseness the sole source of sequential value. We then settle the exact question this raises: a sequential-value gap exists if and only if forced reachability exceeds one-shot securability, a reachability-game winning region decidable in polynomial time, subsuming both boundary cases. We show the underlying forced-reachability problem is complete for P, and that the whole framework embeds into the modal μ-calculus, with betrayal the adversarial box and sequential value a strict fixpoint over its one-forcing-step approximant.

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1 Introduction

A wish, an expressed desire for a transformation of the world, behaves as an operator: it receives a current state and produces, or attempts to produce, a different one. A natural formalization models a wish as a non-deterministic function

$$w\colon S \longrightarrow \mathscr{P}(S), \tag{1}$$

with sequential composition $(w_2 \circ w_1)(s) = \bigcup_{s' \in w_1(s)} w_2(s')$ and identity $w_{\mathrm{id}}(s) = \{s\}$. This is clean and self-consistent. It is also, in a precise sense, nothing new: a map of the form (1) is interchangeable with a binary relation $R_w \subseteq S \times S$, the composition above is relational (Kleisli) composition for the powerset monad, w_{id} is the unit, and the resulting theory proves only facts already true of relations and abstract rewriting systems. Nothing in it distinguishes a wish from an arbitrary transition.

What a wish actually is. The naive model collapses three distinct things into one map. When a wish is granted there is (i) what the granter *could* deliver, the feasible states; (ii) what the wisher *asked for*, the literal articulable content of the wish; and (iii) what the wisher *actually wants*, a preference over outcomes never fully written into the wish. The non-determinism of (1) is precisely the granter's discretion among the feasible states satisfying the literal request, and a wish turns into a monkey's paw when that discretion is exercised against the wisher's true preference. We give each ingredient its own structure and let an adversarial granter resolve the residual freedom.

Relation to prior work, stated plainly. Two of the phenomena this framework expresses are already known, in the theory of misspecified reward. Skalse et al. [1] define a proxy as *unhackable* if increasing proxy return never decreases true return, and show that over rich policy sets only trivial (essentially constant or affine-equivalent) proxies are unhackable. Zhuang and Hadfield-Menell [2] analyze an incomplete proxy supported on a strict subset of the state's utility-bearing attributes and give conditions

under which optimizing any such proxy drives true utility arbitrarily low. Our static results (Section 5) are order-theoretic reformulations of these: we make no claim of novelty for them, and we present them precisely to locate our contribution against this literature and to fix vocabulary. The novel content of the paper is sequential and concerns the interaction of expressibility with the *transitivity* of feasibility, which neither of those works addresses.

Contributions. After fixing the framework (Section 3) and recovering the two static impossibility results (Section 5), we

- introduce the *wishing game* and prove a *single-cell reduction* (Lemma 6.2): the wisher's only effective power is to confine the adversarial granter to one cell of the language;
- establish positional determinacy of optimal wishing on finite spaces (Proposition 6.3);
- prove the *box-ordering lemma* (Lemma 6.4), the single operator inequality $\boxdot_{\mathscr{C}} \subseteq \boxdot_{\mathrm{fine}}$, strict exactly when the language fails to separate successors, and derive from it the paper's structural centerpiece (Proposition 6.5): betrayal and sequential value are the *same* inequality read at the first iterate and at the least fixpoint, both caused by non-separation;
- prove the central theorem (Theorem 7.2): under *transitive* feasibility, a single wish is optimal for every utility if and only if the language is successor-separating, and without separation the one-shot value can fall short of the multi-shot value by the full unit-bounded margin (Corollary 7.3), *with the feasibility relation still transitive*, so the separation is structural and not a scaling artifact;
- refute (Remark 7.5) the natural conjecture that this sequential value is intransitivity of the language quotient, and (Remark 6.6) distinguish it from bisimulation and quotient-MDP abstraction;
- give an exact, decidable characterization of when sequential wishing has any value (Theorem 7.9): a gap exists if and only if forced reachability exceeds one-shot securability, a reachability-game winning region computable in polynomial time, which subsumes both halves of the central theorem;
- determine the exact complexity of the underlying computation (Theorem 8.2): the forced-reachability problem WISH-REACH is P-complete, placing the gap problem in polynomial time (Corollary 8.3);
- exhibit the framework's modal content (Section 9): the Wishverse embeds into the modal μ-calculus, with betrayal the adversarial box and sequential value a strict fixpoint.

The upshot is a single phenomenon, language coarseness, underlying both betrayal and the value of sequential wishing.

2 Related Work

Misspecified objectives. Skalse et al. [1] give the first formal definition of reward hacking and show that unhackability is a very strong condition: because expected return is linear in state-action visit counts, over any policy set containing an open set only trivial proxy pairs are unhackable, so nontrivial *simplification* of a reward function is impossible there, while finite or deterministic policy sets do admit nontrivial unhackable pairs. Zhuang and Hadfield-Menell [2] model a principal whose state has L utility-bearing attributes and an agent given a proxy supported on $J < L$ of them, and prove necessary and sufficient conditions under which optimizing the incomplete proxy makes the principal arbitrarily worse off; they advocate dynamic, interactive re-specification as a remedy. Manheim and Garrabrant [3] taxonomize Goodhart's law into regressional, extremal, causal, and adversarial variants; our adversarial granter

instantiates their adversarial variant. Abel et al. [4] study the expressivity of Markov reward, a question dual to ours of what a fixed specification language can pick out. Our framework reproduces the qualitative content of [1, 2] in a static order-theoretic form (Section 5); our sequential results have no counterpart there.

Rewriting, games, and observational abstraction. The naive model (1) is an abstract rewriting system [5], and the wishing game of Section 6 is a two-player reachability game with a terminal payoff [6] in which the wisher controls an *observation partition* of the state space rather than the state itself. This places the framework next to three established lines: reachability and safety games under partial observation, where a controller acts through an information partition; abstraction refinement in verification, where a coarse partition over-approximates and is refined until a property is decided; and reward misspecification in reinforcement learning, where a proxy is an observation-coarsening of the true objective. Our distinctive object is the strict inclusion between the abstract and concrete controllable-predecessor operators (Lemma 6.4) and its two readings; the precise positioning is that we characterize when one-step observational abstraction is sound for control versus when iterated forcing yields a strictly larger controllable set under adversarial intra-cell choice. We are not aware of this characterization in that literature, though its proofs are elementary.

3 The Preference Wishverse

Definition 3.1 (Preference Wishverse)**.** A *preference Wishverse* is a triple $(S, \preceq, \mathrm{Feas})$ where S is a non-empty set of *states*; $\preceq$ is the wisher's *preference*, a preorder, with $\approx$ the induced indifference and $\prec$ strict preference; and $\mathrm{Feas}\colon S \to \mathscr{P}(S)$ is the *feasibility map*, $\mathrm{Feas}(s)$ being the states a granter could deliver from s. The preorder is *total* if for all x, y either $x \preceq y$ or $y \preceq x$. We write R for the feasibility relation, $sRt \iff t \in \mathrm{Feas}(s)$, and call R *transitive* when $t \in \mathrm{Feas}(s)$ and $u \in \mathrm{Feas}(t)$ imply $u \in \mathrm{Feas}(s)$.

Definition 3.2 (Wish, satisfier set)**.** A *wish* is a goal $G \subseteq S$, the states satisfying the literal request. Fix s and write $F := \mathrm{Feas}(s)$. The *satisfier set* of G at s is $F_G := F \cap G$. The wish is *grantable* at s if $F_G \neq \varnothing$, and *impossible* otherwise.

Remark 3.3 (Recovering the naive model). The satisfier set F_G is exactly the image $w(s)$ of a naive wish (1), but now its non-determinism is explained as the intersection of *what was requested* with *what is feasible*. Impossibility ($F_G = \varnothing$) is the empty-image failed wish of the naive theory, now with a cause.

Throughout we assume relevant extrema exist; this holds for finite F or order-complete $\preceq$.

Definition 3.4 (Granting regimes)**.** For grantable G at s with $F = \mathrm{Feas}(s)$, define $a(G) := \min_{\preceq} F_G$ (the *adversarial* grant, a least feasible satisfier, the monkey's paw), $b(G) := \max_{\preceq} F_G$ (the *benevolent* grant), and the *optimum* $M := \max_{\preceq} F$, with top indifference class $\mathrm{Top}(s) := \{t \in F : t \approx M\}$. Always $a(G) \preceq b(G) \preceq M$. A grantable wish is *betrayal-free* at s if $a(G) \approx b(G)$ and *safe* at s if $a(G) \approx M$; safety implies betrayal-freedom.

4 Instances of the Framework

The genie of folklore is the limiting whimsical case of a structure that recurs wherever an agent issues an under-specified directive to a more capable executor whose interests are not identical to its own. In each instance below, the three ingredients map onto concrete objects and the adversarial granter models the worst-case (or merely indifferent) resolution of what the directive left open.

Reward hacking in reinforcement learning. States S are environment configurations, $\mathrm{Feas}(s)$ the configurations a policy can bring about, the preference $\preceq$ is the designer's true reward, and the language is the level-set partition of the proxy reward: two states share a cell when the proxy cannot distinguish them. A wish is a proxy-reward target; an agent maximizing the proxy is a granter free to deliver any feasible state inside the targeted proxy-cell, and betrayal is its landing on the true-reward-worst member of that cell. This is the setting of Skalse et al. [1] and Zhuang and Hadfield-Menell [2].

Natural-language instruction to an autonomous agent. A user's instruction to an LLM-based agent is a wish: a coarse cell, because language under-determines the intended outcome. The agent is the granter, resolving the ambiguity; the user's intent is the preference. Specification gaming is adversarial choice within the instruction's cell, and the sequential result (Theorem 7.2) says a sequence of instructions can achieve outcomes no single instruction can, even when the agent's feasible repertoire is fixed.

Incomplete legal contracts. The contracted terms partition outcomes into cells (what the contract treats as compliant); the counterparty is the granter, delivering the cheapest compliant outcome; the drafting party's interest is the preference. This is incomplete-contract theory, and Theorem 7.2(b) gives a formal sense in which staged or sequential contracting can extract value a single contract cannot, with the feasible set unchanged.

The genie. States are world-configurations, Feas what the genie can conjure, the preference the wisher's true desire, and the language the wisher's vocabulary. The monkey's paw is the adversarial grant, and the "three wishes" of folklore are a wish budget whose value, by Theorem 7.9, is governed by forced reachability against one-shot securability.

5 Static Theory: Two Recovered Impossibility Results

This section reformulates known facts; see Section 2. The novelty of the paper begins in Section 6.

5.1 Betrayal-freedom and the impossibility of safe coarse wishes

Theorem 5.1 (Safe-wish characterization)**.** *Let $\preceq$ be total and G grantable at s. Then G is safe at s if and only if $F_G \subseteq \mathrm{Top}(s)$.*

Proof. If $F_G \subseteq \mathrm{Top}(s)$ then $a(G) = \min_{\preceq} F_G \approx M$. Conversely if some $t \in F_G$ lies outside $\mathrm{Top}(s)$, then $M \succeq t$ and $t \not\approx M$, so by totality $M \succ t$, giving $a(G) \preceq t \prec M$ and G unsafe. □

Corollary 5.2 (Guaranteed betrayal over coarse languages)**.** *Let wishes be drawn from a language $\mathscr{G} \subseteq \mathscr{P}(S)$. If no $G \in \mathscr{G}$ has $\varnothing \neq F_G \subseteq \mathrm{Top}(s)$, then every grantable $G \in \mathscr{G}$ satisfies $a(G) \prec M$: every expressible wish is betrayable.*

Remark 5.3 (Relation to unhackability). Betrayal-freedom is the order-theoretic analogue of the unhackability of Skalse et al. [1]: a betrayal-free wish is one the granter cannot make worse within the wish, just as an unhackable proxy is one optimization cannot make worse in true return. Corollary 5.2 is the analogue of their impossibility of nontrivial simplification over rich policy sets: a language too coarse to isolate $\mathrm{Top}(s)$ admits no safe wish, exactly as a policy set too rich admits no nontrivial unhackable proxy. The settings differ (their linearity in occupancy measures versus our adversarial choice over a satisfier set), but the qualitative content coincides, and we claim no novelty for it.

5.2 A coarseness bound on betrayal

Let $\preceq$ be represented by a utility $u\colon S \to \mathbb{R}$ and let the language be a Boolean algebra whose atoms partition F into cells $C_1, \dots, C_k$; every expressible F_G is a union of cells. Write $\mathrm{diam}_u(C) := \max_C u - \min_C u$.

Proposition 5.4 (Coarseness bound)**.** *Let $u(M) = \max_F u$ and let $C_\star$ be the cell containing M. Then the betrayal of the best expressible wish obeys*

$$u(M) - \max_G u\big(a(G)\big) \;\le\; \mathrm{diam}_u(C_\star),$$

and a safe expressible wish exists if and only if $C_\star \subseteq \mathrm{Top}(s)$.

Proof. Selecting the cell $C_\star$ guarantees $\min_{C_\star} u = u(M) - \mathrm{diam}_u(C_\star)$, so $\max_G u(a(G)) \ge u(M) - \mathrm{diam}_u(C_\star)$; rearrange. Safety needs $\min_{C_\star} u = u(M)$, i.e. $C_\star \subseteq \mathrm{Top}(s)$. □

Remark 5.5 (Relation to incomplete proxies). This is the bounded, static counterpart of Zhuang and Hadfield-Menell [2]. Their proxy, seeing only $J < L$ attributes, can be driven *arbitrarily* low because their resource-tradeoff geometry lets the agent sacrifice an unmentioned attribute without bound. Our bound is finite precisely because our setting has no such tradeoff: the loss cannot exceed the spread within the forced cell. The shared content is qualitative: a specification coarser than the preference is betrayable, and the betrayal is governed by what the coarsening conflates.

6 The Wishing Game and the Single-Cell Reduction

We now turn to wishing over time, where the contribution lies. Throughout this section and the next, S is finite, preference is a utility $u\colon S \to \mathbb{R}$, and the language is a partition $\mathscr{C}$ of S into *cells*; an expressible wish is a union of cells, and naming a wish G lets the adversarial granter deliver any $t \in \mathrm{Feas}(s) \cap G$. A *pass* keeps the wisher in place.

Definition 6.1 (Wishing game and value)**.** The *t-wish value* is

$$V_0(s) = u(s), \qquad V_t(s) = \max\Big(u(s),\; \max_{\substack{G \text{ expressible} \\ \mathrm{Feas}(s)\cap G \neq \varnothing}} \; \min_{t' \in \mathrm{Feas}(s)\cap G} V_{t-1}(t')\Big).$$

The pass makes V_t non-decreasing in t; on finite S the sequence stabilizes to a limit V_∞ in at most $|S|$ steps.

Lemma 6.2 (Single-cell control)**.** *For every state s and budget, an optimal wish is either the pass or a single cell $C \in \mathscr{C}$ with* $\mathrm{Feas}(s) \cap C \neq \varnothing$.

Proof. An expressible wish is a union of cells $G = \bigcup_i C_i$, so $\mathrm{Feas}(s) \cap G = \bigcup_i (\mathrm{Feas}(s) \cap C_i)$ and, writing $W = V_{t-1}$,

$$\min_{t' \in \mathrm{Feas}(s)\cap G} W(t') \;=\; \min_i \; \min_{t' \in \mathrm{Feas}(s)\cap C_i} W(t') \;\le\; \max_i \; \min_{t' \in \mathrm{Feas}(s)\cap C_i} W(t').$$

The right-hand side is achieved by naming the single best cell C_i with nonempty intersection, which is itself expressible. Enlarging a wish beyond one cell only gives the adversary more to minimize over, so it is never strictly better. □

Lemma 6.2 is the conceptual key: the wisher's only power is to *confine the adversarial granter to one cell*; the granter then moves freely inside $\mathrm{Feas}(s) \cap C$. The entire game is choosing confinements.

Proposition 6.3 (Positional determinacy)**.** *On finite S the limit V_∞ is the least fixpoint above u of the operator in Definition 6.1 and is reached by value iteration in at most $|S|$ steps. An optimal wishing strategy is positional: the optimal wish at any moment depends only on the current state, not on the history of past wishes.*

Proof. The operator is monotone in V and inflationary above u on the finite lattice of attainable value vectors, so iteration from u converges to the least fixpoint in at most $|S|$ steps. The payoff is terminal and the recursion depends only on the current state, so by Lemma 6.2 the maximizing cell at each state is determined by V_∞ at that state, giving a state-indexed optimal policy. □

6.1 The box-ordering lemma

We now isolate the single operator inequality from which both the static betrayal phenomenon and the sequential gap will follow as two readings. In the vocabulary of game solving, the partition $\mathscr{C}$ is an *observation equivalence*: the wisher controls which observation class it names, not which state is delivered, so the language is an information partition and the two operators below are the *concrete* and *abstract controllable-predecessor* operators (CPre) of a two-player reachability game. The *fine* (singleton-control, concrete) operator is the ordinary existential pre-image

$$\boxdot_{\text{fine}} X \;=\; \{\, s : \text{Feas}(s) \cap X \neq \varnothing \,\},$$

modeling a wisher who can name an exact successor; the *coarse* (cell-control, abstract) operator is the controllable pre-image under the observation partition,

$$\boxdot_{\mathscr{C}} X \;=\; \{\, s : \exists C \in \mathscr{C},\ \varnothing \neq \text{Feas}(s) \cap C \subseteq X \,\},$$

modeling a wisher who names an observation class and lets the adversary choose inside it. In these terms the lemma below is the standard fact that the abstract CPre under-approximates the concrete one, with the inclusion strict exactly when the abstraction loses control information.

Lemma 6.4 (Box ordering)**.** *For every $X \subseteq S$, $\boxdot_{\mathscr{C}} X \subseteq \boxdot_{\text{fine}} X$. Moreover the two operators coincide on all X if and only if $\mathscr{C}$ is successor-separating (Definition 7.1).*

Proof. Inclusion. If $s \in \boxdot_{\mathscr{C}} X$ via a cell C, then $\varnothing \neq \text{Feas}(s) \cap C \subseteq X$; any $t \in \text{Feas}(s) \cap C$ witnesses $\text{Feas}(s) \cap X \neq \varnothing$, so $s \in \boxdot_{\text{fine}} X$. The inclusion runs in the direction the adversary's disadvantage predicts: cell control is never stronger than singleton control.

Equality iff separation. If $\mathscr{C}$ is successor-separating, every $t \in \text{Feas}(s)$ is the unique feasible member of its cell $[t]$, so $t \in X$ gives $\text{Feas}(s) \cap [t] = \{t\} \subseteq X$ and hence $s \in \boxdot_{\mathscr{C}} X$; thus $\boxdot_{\text{fine}} X \subseteq \boxdot_{\mathscr{C}} X$, and with the inclusion the two coincide. Conversely, if $\mathscr{C}$ is not successor-separating, some s has distinct $t_1, t_2 \in \text{Feas}(s) \cap C$ for a common cell C; take $X = \{t_1\}$. Then $s \in \boxdot_{\text{fine}} X$ via t_1, but no cell has its feasible image at s inside $\{t_1\}$ (the cell C also carries t_2, and every other cell misses t_1), so $s \notin \boxdot_{\mathscr{C}} X$. The operators differ at $X = \{t_1\}$. □

The next proposition is the structural heart of the paper: betrayal and sequential value are not merely analogous, they are the same operator inequality of Lemma 6.4 read at two points, the first iterate and the least fixpoint. We state it for a total preference, where "the adversary delivers the worse element of a cell" is well defined; the box ordering itself is purely set-theoretic and needs no preference.

Proposition 6.5 (One inequality, two effects)**.** *Let the preference be total. For a target set T write $\Psi_{\text{fine}}(X) = T \cup \boxdot_{\text{fine}} X$ and $\Psi_{\mathscr{C}}(X) = T \cup \boxdot_{\mathscr{C}} X$. Then:*

(i) (Betrayal, first iterate) $\Psi_{\mathscr{C}}(T) \subsetneq \Psi_{\text{fine}}(T)$ *for some* T *if and only if some wish is betrayable, i.e. some state has a feasible non-optimal satisfier sharing a cell with an optimal one;*

(ii) (Sequential value, fixpoint) $\mu X.\Psi_{\mathscr{C}}(X) \subsetneq \mu X.\Psi_{\text{fine}}(X)$ *for some* T *if and only if a sequential-value gap exists.*

Both strictness conditions hold precisely when $\mathscr{C}$ *is not successor-separating, so both effects have the single common cause of Lemma 6.4.*

Proof. By Lemma 6.4, $\boxdot_{\mathscr{C}} = \boxdot_{\text{fine}}$ on all X iff $\mathscr{C}$ is successor-separating; otherwise they differ at some $X = \{t_1\}$.

(i) If $\mathscr{C}$ is not separating, the witnessing s has feasible t_1, t_2 in one cell C; taking T to contain the $\preceq$-larger of them and not the smaller, the adversary confined to C delivers the smaller, which is a betrayable wish at s (the static notion of Section 5). If $\mathscr{C}$ is separating then $\boxdot_{\mathscr{C}} T = \boxdot_{\text{fine}} T$ for every T, so no first-iterate strictness and, by single-cell control, no cell hides a worse satisfier: no betrayal.

(ii) If $\mathscr{C}$ is separating the two operators are identical, hence so are their least fixpoints, and there is no gap. If not, monotonicity gives $\mu\Psi_{\mathscr{C}} \subseteq \mu\Psi_{\text{fine}}$, and Theorem 7.9 provides a target witnessing strict inclusion exactly when $\text{Reach}_\infty \nsubseteq \text{Sec}_1$, i.e. a gap. The common cause is Lemma 6.4 in both cases. □

Remark 6.6 (Why this is not bisimulation or quotient-MDP abstraction). The controllable pre-image $\boxdot_{\mathscr{C}}$ is *not* the transition operator of a quotient structure under bisimulation. Successor-separation is strictly weaker than bisimulation: it requires each state's successors to fall in distinct cells, but does not require two states in a common cell to have the same successor-cells. Hence there are non-bisimulation partitions that are nonetheless successor-separating, on which, by Proposition 6.5, there is no gap. The gap condition of Theorem 7.9 is therefore strictly finer than bisimulation failure, and the phenomenon is invisible to standard quotient-MDP abstraction, whose relevant equivalence is bisimulation. What the adversarial-granting model detects is precisely the regime of coarse one-shot control with intact forced reachability, the strict gap $\boxdot_{\mathscr{C}} \subsetneq \boxdot_{\text{fine}}$.

7 Coarseness, Not Intransitivity, Governs Sequential Value

For the naive (singleton-fine) language, transitivity of R forces one wish to be optimal: a wisher who can name any single state, applied to a relation where every reachable state is already directly feasible, has nothing to gain from a second wish. We are precise about what transitivity does and does not do: a transitive R equals its own transitive closure, so the *concrete* reachable set already saturates under one step, but, as the theorem shows, the *abstract* controllable-reachable set under a coarse observation partition need not. Transitivity is imposed here deliberately, to neutralize the ordinary source of sequential value, the existence of long forced paths in the concrete relation, so that any sequential value that remains can have only one explanation, the coarseness of the observation partition. The theorem below is therefore not surprising in its first half (under a fine partition, concrete saturation transfers to control) but in its second: sequential value survives precisely where the only available cause has been assumed away. We measure the gap with bounded utilities, so the separation is structural rather than an artifact of scaling.

Definition 7.1 (Successor-separating language)**.** A partition $\mathscr{C}$ is *successor-separating* if $|\text{Feas}(s) \cap C| \leq 1$ for every $s \in S$ and $C \in \mathscr{C}$, that is, no state has two distinct feasible successors in a common cell.

Successor-separation is the precise sense in which the language is fine enough to aim: naming the cell of a successor then forces exactly that successor, restoring full singleton control over $\text{Feas}(s)$.

Theorem 7.2 (Coarseness governs sequential value)**.** *Let* R *be transitive.*

(a) *If $\mathscr{C}$ is successor-separating, then $V_1 = V_\infty$: one wish is optimal at every state and for every utility u.*

(b) *If $\mathscr{C}$ is not successor-separating, then* (a) *can fail even with utilities bounded in $[0,1]$: there is a transitive R and a non-successor-separating $\mathscr{C}$ carrying a $\{0,1\}$-valued utility for which $V_\infty - V_1 = 1$, the maximum separation a unit-bounded utility allows.*

Proof. **(a)** Successor-separation means each $t \in \mathrm{Feas}(s)$ is the unique element of $\mathrm{Feas}(s) \cap [t]$, so naming $[t]$ forces delivery of t exactly: the wisher has full singleton control over $\mathrm{Feas}(s)$. Transitivity makes every state reachable from s already lie in $\mathrm{Feas}(s)$ (any path $s \to x \to \cdots \to z$ composes, by transitivity, to a direct edge $s \to z$). Hence no sequence of wishes can reach beyond $\{s\} \cup \mathrm{Feas}(s)$, so $V_\infty(s) \le \max\big(u(s), \max_{t \in \mathrm{Feas}(s)} u(t)\big)$. The wisher attains this bound in one wish by naming the cell of an optimal successor (or passing), so $V_1(s) = V_\infty(s)$.

(b) Take $S = \{s_0, a, b, g\}$ with $R = \{(s_0,a), (s_0,b), (s_0,g), (a,g)\}$. This R is transitive: the only composable path is $s_0 \to a \to g$, and its shortcut (s_0,g) is present. Let the cells be $\{s_0\}$, $\{b\}$, $\{a,g\}$; the cell $\{a,g\}$ violates successor-separation at s_0, since $\mathrm{Feas}(s_0) \cap \{a,g\} = \{a,g\}$. Put $u(g) = 1$ and $u = 0$ elsewhere, so u is $\{0,1\}$-valued.

One wish from s_0. The grantable cells meeting $\mathrm{Feas}(s_0) = \{a,b,g\}$ are $\{b\}$, giving 0, and $\{a,g\}$, where the granter delivers the worse of a, g, namely a, giving 0. So $V_1(s_0) = 0$.

Two wishes from s_0. Name $\{a,g\}$; the granter is forced to a (utility 0). From a we have $\mathrm{Feas}(a) = \{g\}$, so naming $\{a,g\}$ at a leaves the granter only g, giving $V_1(a) = 1$. Hence $V_2(s_0) = 1 > 0 = V_1(s_0)$, so $V_\infty(s_0) - V_1(s_0) = 1$, the largest gap a $[0,1]$-valued utility permits. □

Corollary 7.3 (One-shot inapproximability)**.** *For every $\alpha < 1$ there is a transitive Wishverse with utilities in $[0,1]$ and a state s at which $V_1(s) \le \alpha V_\infty(s)$. Consequently no single wish, and indeed no number of wishes below the forced-reachability depth, achieves a constant-factor approximation of the optimal guaranteed value in general.*

Proof. Chain k copies of the gadget of part (b) so that the unique value-1 state lies at forced-reachability depth k from s and is reachable only through k adversarially forced stepping stones, each one-shot-securing value 0; then $V_1(s) = 0$ while $V_\infty(s) = 1$, giving $V_1(s) \le \alpha V_\infty(s)$ for any α once $V_1(s) = 0$. A smoothed variant placing small positive utilities on the intermediate stones realizes any target ratio $\alpha < 1$ with $O(1/(1-\alpha))$ states. □

Corollary 7.4 (Necessity of coarseness)**.** *Under transitive feasibility, any positive gap $V_\infty - V_1$ at any state, for any utility, implies the language is not successor-separating. Equivalently, with a successor-separating language the value of sequential wishing over a transitive feasibility relation is zero for every utility. We emphasize the quantifier: successor-separation is necessary and sufficient for "no gap for* every *utility," which is the level at which Theorem 7.2(a) and this corollary form an equivalence; for a single fixed utility a gap may be absent without separation, and the exact per-instance condition is Theorem 7.9.*

Proof. Contrapositive of Theorem 7.2(a). □

Remark 7.5 (Refutation of the quotient-intransitivity conjecture). One expects the sequential value in Theorem 7.2(b) to be intransitivity of the language quotient. It is not. Lift R to cells by $\widehat{R}([s]) = \{[t] : t \in \mathrm{Feas}(s)\}$. In the example $\widehat{R}$ contains $[s_0] \to \{a,g\}$ and $\{a,g\} \to \{a,g\}$ and is transitive, yet the gap is positive. Moreover a "singleton wish on the quotient" is not even a forced move, because naming the cell $\{a,g\}$ still leaves the granter a choice within it. The correct mechanism is the granter's *freedom inside a coarse cell*: minimizing utility, the adversary hands the wisher a rather than g, but a has strictly better onward reachability, so the short-run betrayal is exactly what makes the second wish productive. Betrayal

(Section 5) and the value of sequential wishing (Theorem 7.2) are two faces of one cause: language coarseness.

7.1 The exact gap characterization

Theorem 7.2 gives a sufficient condition for no gap (separation) and, via Corollary 7.4, a necessary one (non-separation), but not an equivalence: non-separation alone does not force a gap. We now close this with an exact, decidable characterization. It turns out the natural local guess, that a gap exists whenever some confining cell holds two successors one of which forcibly reaches higher, is too shallow: the productive stepping stone may itself require a multi-wish detour, so the condition must be a fixpoint, not a two-step test. The right objects are a reachability winning region and a one-shot secure set.

Definition 7.6 (One-shot securability)**.** A state t is *one-shot securable* at s if some cell C has $\mathrm{Feas}(s)\cap C = \{t\}$: the wisher can name a cell that forces exactly t. Write

$$\mathrm{Sec}_1(s) \;:=\; \{s\}\cup\{\,t : \exists C\in\mathscr{C},\ \mathrm{Feas}(s)\cap C=\{t\}\,\}.$$

Definition 7.7 (Forced reachability)**.** A wish at s *forces* the wisher into a target $T\subseteq S$ if some cell C has $\varnothing\neq\mathrm{Feas}(s)\cap C\subseteq T$: every state the adversary may deliver lies in T. The *forced-reachability set* $\mathrm{Reach}_\infty(s)$ is the least fixpoint of

$$\mathrm{Reach}_0(s)=\{s\},\qquad \mathrm{Reach}_{k+1}(s)=\mathrm{Reach}_k(s)\cup\Big\{\,t : \exists r\in\mathrm{Reach}_k(s),\ \exists C\in\mathscr{C},\ \varnothing\neq\mathrm{Feas}(r)\cap C\subseteq\mathrm{Reach}_k(s)\cup\{t\}\,\Big\},$$

reached in at most $|S|$ steps. Equivalently, $t\in\mathrm{Reach}_\infty(s)$ iff the wisher has a positional strategy guaranteeing arrival at t from s against every adversary; this is the winning region of the finite reachability game in which the wisher names cells and the adversary chooses within $\mathrm{Feas}\cap C$ [6].

For the characterization we need the precise sense in which Reach_∞ is a winning region. For a fixed target g, define the backward attractor $\mathrm{Attr}_0(g)=\{g\}$, $\mathrm{Attr}_{k+1}(g)=\mathrm{Attr}_k(g)\cup\{\,r : \exists C\in\mathscr{C},\ \varnothing\neq\mathrm{Feas}(r)\cap C\subseteq\mathrm{Attr}_k(g)\,\}$, and $\mathrm{Attr}_\infty(g)=\bigcup_k\mathrm{Attr}_k(g)$. This is the dual of Definition 7.7: $g\in\mathrm{Reach}_\infty(s)$ iff $s\in\mathrm{Attr}_\infty(g)$, both expressing that the wisher can force the play from s to g.

Lemma 7.8 (Forced reachability is a winning region)**.** *For all states s,g: the wisher has a strategy guaranteeing arrival at g from s if and only if $s\in\mathrm{Attr}_\infty(g)$, equivalently $g\in\mathrm{Reach}_\infty(s)$. Consequently, if $g\notin\mathrm{Reach}_\infty(s)$ then the adversary has a positional strategy preventing arrival at g from s forever.*

Proof. ($\Leftarrow$) Suppose $s\in\mathrm{Attr}_\infty(g)$, and let k be least with $s\in\mathrm{Attr}_k(g)$. We induct on k that the wisher forces arrival at g from s. If $k=0$ then $s=g$ and arrival is immediate. If $k>0$, then s entered at stage k because some cell C has $\varnothing\neq\mathrm{Feas}(s)\cap C\subseteq\mathrm{Attr}_{k-1}(g)$. The wisher names C; every state the adversary may deliver lies in $\mathrm{Feas}(s)\cap C\subseteq\mathrm{Attr}_{k-1}(g)$, hence has attractor-rank at most $k-1$, so by the induction hypothesis the wisher forces g from each. The strategy "name C, then follow the rank-$(k-1)$ strategy from wherever the adversary delivers" forces g from s, with the rank strictly decreasing each step and therefore terminating.

($\Rightarrow$) Contrapositive. Suppose $s\notin\mathrm{Attr}_\infty(g)$, and set $B:=S\setminus\mathrm{Attr}_\infty(g)$, so $s\in B$ and $g\notin B$ (as $g\in\mathrm{Attr}_0(g)$). We show B is a *trap*: for every $r\in B$ and every cell C with $\mathrm{Feas}(r)\cap C\neq\varnothing$, we have $\mathrm{Feas}(r)\cap C\cap B\neq\varnothing$. Indeed, if instead $\mathrm{Feas}(r)\cap C\subseteq\mathrm{Attr}_\infty(g)$ for some grantable C, then r would satisfy the attractor step over the fixpoint $\mathrm{Attr}_\infty(g)$ and hence lie in it, contradicting $r\in B$. So whatever grantable cell the wisher names at a state of B, the positional adversary rule "deliver some member of $\mathrm{Feas}(r)\cap C\cap B$" keeps the play in B. Since $s\in B$ and $g\notin B$, this strategy prevents arrival at g forever. □

Theorem 7.9 (Exact gap characterization)**.** *Let R be transitive and $\mathscr{C}$ a partition of the finite set S. The following are equivalent.*

(1) *There exist a utility u and a state s with $V_\infty(s) > V_1(s)$, i.e. a sequential-value gap exists.*

(2) *There is a state s with $\mathrm{Reach}_\infty(s) \not\subseteq \mathrm{Sec}_1(s)$: forced reachability exceeds one-shot securability.*

Proof. $(2 \Rightarrow 1)$ Fix s and $g \in \mathrm{Reach}_\infty(s) \setminus \mathrm{Sec}_1(s)$. Put $u(g) = 1$ and $u = 0$ elsewhere. Since g is forcibly reachable, the wisher has a strategy guaranteeing arrival at g, so $V_\infty(s) \geq 1$. For the one-shot value, by single-cell control (Lemma 6.2) any single wish names a cell C and yields $\min_{t \in \mathrm{Feas}(s) \cap C} u(t)$. If $g \notin \mathrm{Feas}(s) \cap C$ this minimum is 0. If $g \in \mathrm{Feas}(s) \cap C$ then, because $g \notin \mathrm{Sec}_1(s)$, the cell C contains another feasible state $t \neq g$ with $u(t) = 0$, so again the minimum is 0. Hence $V_1(s) = 0 < 1 \leq V_\infty(s)$.

$(1 \Rightarrow 2)$ Contrapositive: assume $\mathrm{Reach}_\infty(s) \subseteq \mathrm{Sec}_1(s)$ for every s, and fix any utility u and any state s. In a terminal-payoff reachability game the wisher's guaranteed value is $V_\infty(s) = \max\{u(g) : g \in \mathrm{Reach}_\infty(s)\}$: the wisher can secure value $u(g)$ exactly by forcing arrival at g, which by Lemma 7.8 is possible iff $g \in \mathrm{Reach}_\infty(s)$, and for any $g \notin \mathrm{Reach}_\infty(s)$ the adversary prevents arrival, so no value above $\max_{g \in \mathrm{Reach}_\infty(s)} u(g)$ can be guaranteed. By hypothesis $\mathrm{Reach}_\infty(s) \subseteq \mathrm{Sec}_1(s)$, so this maximum is attained over $\mathrm{Sec}_1(s)$, every element of which is one-shot securable (or s itself), hence achievable in a single wish. Therefore $V_1(s) = \max_{g \in \mathrm{Sec}_1(s)} u(g) = V_\infty(s)$. As u and s were arbitrary, no gap exists. □

Remark 7.10 (The operator-level invariant behind necessity). The reason successor-separation is the exact boundary is algebraic, not incidental. By Lemma 6.4 the abstract and concrete controllable-predecessor operators coincide on all sets iff $\mathscr{C}$ is successor-separating; when they coincide their least fixpoints agree, so $\mathrm{Reach}_\infty = \mathrm{Sec}_1$-style one-step control everywhere and no utility can separate V_1 from V_∞; when they differ, Theorem 7.9 exhibits the separating utility. Thus the "for every utility" necessity of Corollary 7.4 is precisely the operator strictness $\boxdot_{\mathscr{C}} \subsetneq \boxdot_{\mathrm{fine}}$, which is the deeper invariant the per-instance condition refines.

Corollary 7.11 (Both parts of Theorem 7.2 as special cases)**.** *If $\mathscr{C}$ is successor-separating then* $\mathrm{Sec}_1(s) = \{s\} \cup \mathrm{Feas}(s)$*, and transitivity gives* $\mathrm{Reach}_\infty(s) = \{s\} \cup \mathrm{Feas}(s)$*, so condition* (2) *fails everywhere and there is no gap, recovering Theorem 7.2*(a)*. The four-state Wishverse of Theorem 7.2*(b) *is the minimal witness of* (2)*: $g \notin \mathrm{Sec}_1(s_0)$ because the only cell containing g also contains a, yet $g \in \mathrm{Reach}_\infty(s_0)$ via the forced step to a then to g, so* $\mathrm{Reach}_\infty(s_0) \not\subseteq \mathrm{Sec}_1(s_0)$.

Remark 7.12 (Decidability and complexity). Both Sec_1 and Reach_∞ are computed by the finite iterations of Definitions 7.6 and 7.7: Sec_1 in a single pass over states and cells, and Reach_∞ as a least-fixpoint reachability computation converging in at most $|S|$ rounds. Hence whether a Wishverse admits any sequential value is decidable in polynomial time, of order $O(|S| \cdot |\mathscr{C}| \cdot |R|)$ in the obvious representation. The characterization is thus not only exact but effective.

Remark 7.13 (Self-containment). The converse does not depend on any external determinacy theorem. Lemma 7.8 is proved directly: the $(\Leftarrow)$ direction by rank induction on the backward attractor, and the $(\Rightarrow)$ direction from the elementary observation that the complement of a least attractor is a trap. By single-cell control (Lemma 6.2) the wisher's moves are exactly "name a cell," so the wishing game is a finite reachability game whose winning region is Reach_∞ by construction. The characterization is therefore elementary and self-contained.

8 Complexity of the Wishing Game

The characterization of Section 7.1 is effective, and we now pin down its complexity exactly. The computational heart is not the gap question itself but the forced-reachability problem it calls, which we

isolate and show complete for P. We drop transitivity here: forced reachability is well defined for any finite R, and the result is cleanest stated in that generality.

Definition 8.1 (WISH-REACH)**.** WISH-REACH is the decision problem: given a finite Wishverse with partition $\mathscr{C}$, a source s, and a target t, is $t \in \mathrm{Reach}_\infty(s)$, that is, can the wisher force arrival at t against every adversary?

Theorem 8.2 (WISH-REACH is P-complete)**.** WISH-REACH *is complete for* P *under logspace many-one reductions.*

Proof. Membership. The forced-reachability set is the attractor fixpoint of Definition 7.7, computed by monotone iteration converging in at most $|S|$ rounds; each round scans states and cells, giving total time $O(|S| \cdot |\mathscr{C}| \cdot |R|)$, hence membership in P.

Hardness. We reduce from the Monotone Circuit Value Problem (MCVP), which is P-complete under logspace reductions [7]. A monotone circuit has AND and OR gates and constant TRUE/FALSE inputs; the question is whether the designated output gate evaluates to TRUE. We encode the circuit as the standard alternating-reachability game graph: OR-gates become *controller* (wisher) positions, where the existential choice of a successor is the wisher's, and AND-gates become *adversary* positions, where the universal choice is the granter's. The observation partition realizes this split, distinct cells at a position give the wisher the choice (controller), a shared cell forces the wisher to offer both and hands the choice to the adversary, so that forcing the target reproduces monotone circuit evaluation.

For each gate g introduce a state p_g in its own singleton cell; add a target $\top$ and a sink $\bot$, each in its own cell, with $\mathrm{Feas}(\top) = \{\top\}$ and $\mathrm{Feas}(\bot) = \{\bot\}$. Every wire is routed through a fresh *port* state, with $\mathrm{Feas}(e) = \{p_h\}$ for the port e on the wire into child p_h, and p_h in its own singleton cell so that the port forces p_h in one step. The gadgets are:

- TRUE input g: $\mathrm{Feas}(p_g) = \{\top\}$.
- FALSE input g: $\mathrm{Feas}(p_g) = \{\bot\}$.
- OR gate $g = g_1 \vee g_2$: ports e_1, e_2 with $\mathrm{Feas}(p_g) = \{e_1, e_2\}$ and e_1, e_2 in *different* cells.
- AND gate $g = g_1 \wedge g_2$: ports e_1, e_2 with $\mathrm{Feas}(p_g) = \{e_1, e_2\}$ and e_1, e_2 in the *same* cell.

All gate-states and the constants get singleton cells; the only same-cell and different-cell constraints fall on ports, which are never shared between gates, so the cells form a valid partition and the construction is computable in logarithmic space by local transcription.

Write $A = \mathrm{Reach}_\infty^{-1}(\{\top\})$ for the attractor of $\{\top\}$, the set of states from which the wisher forces $\top$; recall $r \in A$ iff some cell C has $\varnothing \neq \mathrm{Feas}(r) \cap C \subseteq A$, with $\top \in A$. We first record the port fact, then the gate induction.

Port fact. For a port e with $\mathrm{Feas}(e) = \{p_h\}$ and p_h singleton-celled, $e \in A$ iff $p_h \in A$: if $p_h \in A$ then naming $\{p_h\}$ forces A, so $e \in A$; conversely e's only feasible state is p_h, so $e \in A$ requires $\{p_h\} \subseteq A$.

Gate induction on circuit depth, claim: $p_g \in A$ iff g evaluates to TRUE. *Base.* A TRUE input has $\mathrm{Feas}(p_g) = \{\top\} \subseteq A$ via $\top$'s cell, so $p_g \in A$. A FALSE input has $\mathrm{Feas}(p_g) = \{\bot\}$; since $\mathrm{Feas}(\bot) = \{\bot\}$ and $\bot \neq \top$, the value iteration never places $\bot$ in A, so $p_g \notin A$. These match the constants. *Step,* OR. The cells of e_1, e_2 differ, so $p_g \in A$ holds as soon as some $\{e_i\} \subseteq A$, i.e. iff $e_1 \in A$ or $e_2 \in A$, iff (port fact) $p_{g_1} \in A$ or $p_{g_2} \in A$, iff (induction hypothesis) g_1 or g_2 is TRUE, iff g is TRUE. *Step,* AND. The ports e_1, e_2 share a cell C, and it is the only cell meeting $\mathrm{Feas}(p_g) = \{e_1, e_2\}$, with $\mathrm{Feas}(p_g) \cap C = \{e_1, e_2\}$; hence the forcing condition $\mathrm{Feas}(p_g) \cap C \subseteq A$ holds iff both $e_1, e_2 \in A$, iff (port fact) both $p_{g_1}, p_{g_2} \in A$, iff (induction hypothesis) both g_1, g_2 TRUE, iff g is TRUE. This closes the induction.

Therefore $p_{\text{out}} \in A$, equivalently $\top \in \text{Reach}_\infty(p_{\text{out}})$, iff the circuit outputs TRUE, which is the required logspace reduction. □

Because alternating reachability is P-hard, WISH-REACH is not in NL unless NL = P; the P bound is tight. The gap problem inherits a polynomial algorithm.

Corollary 8.3 (WISH-GAP is polynomial)**.** *For transitive R, the problem of deciding whether a sequential-value gap exists, that is whether some s has* $\text{Reach}_\infty(s) \not\subseteq \text{Sec}_1(s)$ *(equivalently, by Theorem 7.9, whether some utility produces* $V_\infty > V_1$*), is solvable in polynomial time.*

Proof. Compute $\text{Sec}_1(s)$ in one pass and $\text{Reach}_\infty(s)$ by Theorem 8.2's attractor for each s, then test inclusion; all steps are polynomial. □

Remark 8.4 (On the hardness of WISH-GAP itself). We do not claim WISH-GAP is P-complete. The "does a gap exist anywhere" phrasing is satisfied by many structurally trivial instances, so it is a poor target for a hardness reduction, and whether it admits one, particularly under the transitivity restriction, is left open. The robust complexity statement is Theorem 8.2 for the underlying reachability problem, of which Corollary 8.3 is a consequence.

9 Embedding into the Modal μ-Calculus

The wishing game's objects are modal, and making the translation explicit connects the framework to dynamic logic [10] and locates the gap as a fixpoint phenomenon. We read state-properties as propositions and each cell as a parameter of an adversarial box.

Definition 9.1 (Granting modalities)**.** For a cell $C \in \mathscr{C}$ and a proposition φ with extension $[\![\varphi]\!] \subseteq S$, define the *adversarial box*

$$s \vDash [C]\varphi \quad :\Longleftrightarrow \quad \varnothing \neq \text{Feas}(s) \cap C \subseteq [\![\varphi]\!],$$

"naming C is grantable and guarantees φ whatever the granter delivers."

The adversarial granter is exactly why the box, not the diamond, is the operative modality: the wisher controls the cell but not the choice within it. Single-cell control (Lemma 6.2) is now a modal identity.

Proposition 9.2 (Single-cell control, modally)**.** *Let the wisher's one-step guaranteed modality be* $\boxdot\varphi := \bigvee_{C\in\mathscr{C}}[C]\varphi$*. Then* $\boxdot$ *is the set of propositions the wisher can secure in one wish, and enlarging a wish beyond a single cell never helps: for cells* C_1, C_2*,* $[C_1 \cup C_2]\varphi$ *implies* $[C_1]\varphi \vee [C_2]\varphi$ *fails in general, yet the securable set is still* $\bigvee_C [C]\varphi$*, attained at a single cell.*

Proof. Immediate from Lemma 6.2: the best one-wish guarantee is the disjunction over single cells of the adversarial box, and a union of cells only enlarges the set the granter minimizes over. □

Proposition 9.3 (Forced reachability is a least fixpoint, and the gap is approximant strictness)**.** *Fix a target state t and the target proposition* $T = \{t\}$*. Let* $\Phi_T := \mu X.\,(T \vee \boxdot X) = \mu X.\,\big(T \vee \bigvee_{C\in\mathscr{C}}[C]X\big)$*, with Knaster-Tarski approximants* $X_0 = \varnothing$*,* $X_{k+1} = T \vee \boxdot X_k$*. Then:*

(i) $[\![\Phi_T]\!] = \{s :\ t \in \text{Reach}_\infty(s)\}$*: the extension of the fixpoint is exactly the set of states from which the wisher forces arrival at t;*

(ii) *the second approximant* $X_2 = [\![T \vee \boxdot T]\!]$ *is the set of states from which t is forced in at most one wish;*

(iii) *a sequential-value gap exists if and only if, for some target t,* $[\![\Phi_T]\!] \supsetneq X_2$*: the fixpoint strictly exceeds its one-forcing-step approximant.*

Proof. The operator $\Psi(X) = T \vee \boxdot X$ is monotone (the box $[C]$ is monotone in its argument and disjunction preserves monotonicity), so by Knaster-Tarski its least fixpoint is $\bigcup_k X_k$ with $X_{k+1} = \Psi(X_k)$. By definition of $\boxdot$ as the disjunction of adversarial boxes over cells, X_{k+1} is the set of states from which the wisher forces t within k wishes, so the union is $\{s : t \in \mathrm{Reach}_\infty(s)\}$, giving (i); the case $k = 1$ gives (ii). For (iii), $V_\infty(s) > V_1(s)$ for some utility iff (Theorem 7.9) some s has a forced target beyond one-shot reach, i.e. some t and s with $s \in [\![\Phi_T]\!] \setminus X_2$, which is exactly strictness of the fixpoint over its second approximant. □

Remark 9.4 (Sec_1 versus the approximant). One must not conflate the one-forcing-step approximant X_2 with one-shot securability Sec_1. The set X_2 (relative to target t) collects states from which t is *forced* in one wish, whereas $\mathrm{Sec}_1(s)$ collects states the language can *isolate* from s by a singleton-meeting cell. These coincide exactly when the relevant cells are singletons on $\mathrm{Feas}(s)$, that is under successor-separation; in general X_2 is the larger, and it is the object the value recursion V_1 tracks. This is why coarseness, which separates Sec_1 from the genuine one-step forcing power, is what opens the gap.

This is a reframing rather than a new modal theorem: it places the Wishverse inside the modal μ-calculus, identifies betrayal with the adversarial box and sequential value with fixpoint unfolding, and explains why the whimsical and the formal coincide. The identity wish is the trivial modality, an impossible wish the inconsistent box, and the gap a strict fixpoint, all standard once the embedding is fixed.

10 Discussion

The framework separates the three ingredients the naive wish conflated and assigns each its role. *Preference* and *expressibility* together produce betrayal, and the static results of Section 5 show this reproduces, in order-theoretic form, the known impossibility of betrayal-free coarse specification [1, 2]. *Feasibility* and *expressibility* together produce the sequential phenomenon of Section 7: over a transitive feasibility relation, sequential wishing is worthless under a fine language and can be unboundedly valuable under a coarse one. The unifying point, made precise by Remark 7.5, is that both effects have the same cause. Coarseness is what gives the adversarial granter room to betray, and that same room is what can force the granter to deposit the wisher on a better stepping stone. This is why the natural quotient-intransitivity explanation fails: the relevant object is not how cells relate to cells, but how much freedom a cell leaves the granter at a single step. We state the scope of this claim precisely: *under static transitive feasibility, a fixed observation partition, and deterministic terminal utilities*, observational coarseness is the sole remaining source of sequential value. We make no claim beyond this regime; in general games sequential value also arises from stochastic transitions, path-dependent feasibility, non-commuting actions, and richer partial observation, all of which our modeling choices hold fixed precisely so that the coarseness mechanism can be isolated.

Two consequences are worth drawing out. First, for the design of specifications the result is a caution against the intuition that coarsening is merely lossy: under sequential interaction with an adversarial or indifferent grantor, coarsening changes the reachable set, sometimes favorably, sometimes not, and the sign is governed by Theorem 7.9 (forced reachability versus one-shot securability) rather than by any monotone notion of information. Second, the connection to [2] sharpens: where they show interactive re-specification can help a principal facing an incomplete proxy, our Theorem 7.2(b) isolates a mechanism by which sequential interaction helps even with the *language held fixed*, provided the language is coarse. The help comes not from refining the specification but from the granter's forced moves inside the existing coarse cells.

The honest status is this. The gap question is now settled exactly and effectively (Theorem 7.9, Remark 7.12), subsuming both halves of Theorem 7.2, and the argument is self-contained (Remark 7.13), but

the proofs throughout are elementary. The contribution is the mechanism, the coarseness-breaks-transitivity statement and its exact reachability-game characterization, the P-completeness of the underlying reachability problem, and the recovery of [1, 2] in a common order-theoretic setting, rather than technical depth.

10.1 Limitations

The results rest on four modeling choices, each a genuine restriction. *Finite state space.* Theorem 7.9, Theorem 8.2, and positional determinacy all use finiteness, through the convergence of the attractor iteration and the value recursion. The infinite case requires directed-complete or domain-theoretic structure for the relevant suprema and fixpoints, and the polynomial-time claim has no meaning there. *Adversarial granter.* We resolve the residual non-determinism by worst-case choice. This is the conservative model of a granter whose interests are opposed or merely indifferent; a stochastic granter would replace the inner minimum by an expectation, and a bounded-malice granter by a constrained minimum, changing both the coarseness bound (Proposition 5.4) and the gap condition. *Deterministic utilities and total preference in the sequential theory.* The sequential results take preference as a utility $u\colon S \to \mathbb{R}$, hence a total order. Partial preferences, where the adversary's choice within a cell can be between incomparable optima, are not treated here and would require replacing the minimum with a greatest-lower-bound in a lattice. *Partition language.* We take the specification language to be a partition of S into cells. A general Boolean or sub-lattice language, or a language varying with the state, would refine the notion of one-shot securability and may alter Theorem 7.9; we have not pursued this. The natural next step is to relax the first of these, extending from terminal payoffs to discounted accumulation over an infinite horizon, where the single-cell reduction (Lemma 6.2) should still hold but the characterizing condition will change.

11 Conclusion

Modeled as a non-deterministic map, a wish is a relation and its theory is abstract rewriting. Modeled with feasibility, preference, and expressibility separated, and with an adversarial granter resolving the residual freedom, a wish becomes the carrier of a misspecification phenomenon. In this setting the known impossibility of betrayal-free coarse specification reappears as an order-theoretic fact, and a genuinely new result emerges: over a transitive feasibility relation, a single wish suffices exactly when the language separates successors, and otherwise sequential wishing is unboundedly more valuable, driven not by any quotient intransitivity but by the granter's freedom inside coarse cells. Betrayal and the value of sequential wishing are one phenomenon seen twice. The exact gap condition is settled: a sequential-value gap exists precisely when forced reachability outruns one-shot securability, a reachability-game winning region decidable in polynomial time.

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